

Calculus

Chapter 1: Limits and Continuity

Caitlin Olson
Honors Project

8 - 4 = 6

8 - 4 = 6

8 - 4 = 6

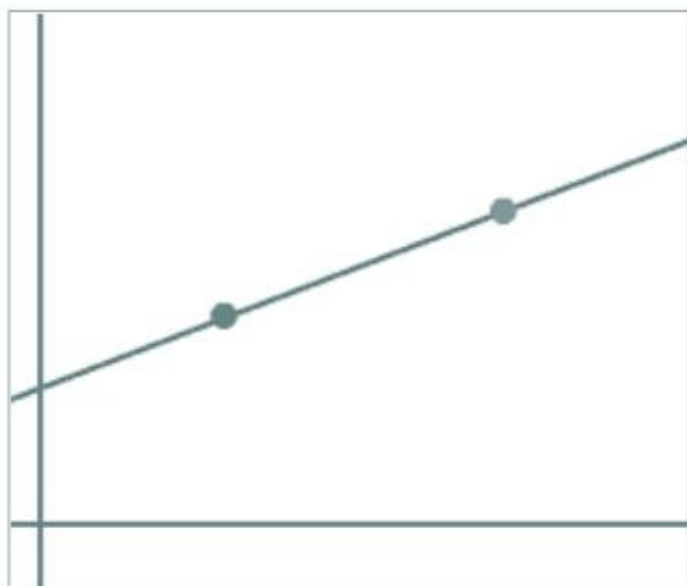
8 - 4 = 6

What is Calculus?

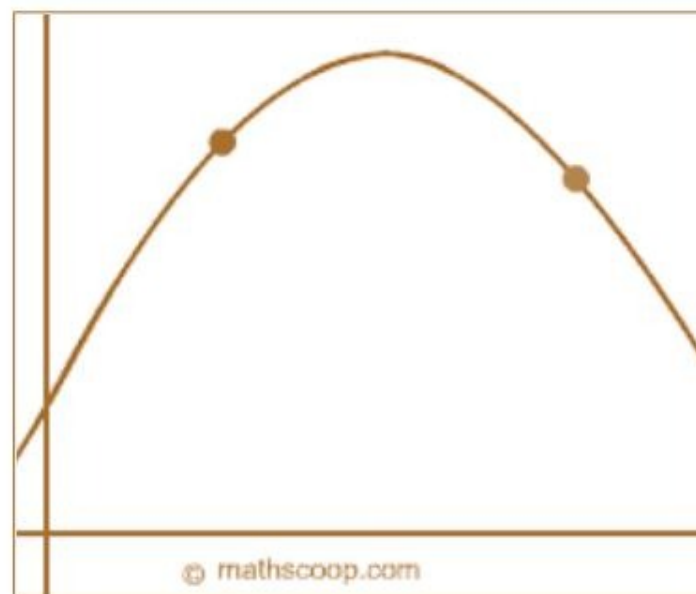
- Calculus is a branch of mathematics that studies change in two branches:
 - Rate of change
 - Integration/Accumulation
- Calculus focuses on limits, functions, derivatives, integrals, and infinite series.
- Calculus uses concepts from Algebra and Geometry and extends on them with limits.
- When Algebra alone is insufficient in solving a problem, you can use Calculus.

Differential Calculus

- This branch of calculus focuses on rate of change.



In Algebra, you would find the slope of this line.



In Calculus, you would find the slope of this curve.

Differential Calculus: Real World Applications

- Differential Calculus is found in fields such as economics, biology, engineering, and physics.
- Differential Calculus can be used to describe exponential growth and decay.
 - Half-lives of radioactive isotopes
 - Population growth
 - Change in investment over time
 - Growth rate of tumors
 - Rates of chemical reactions

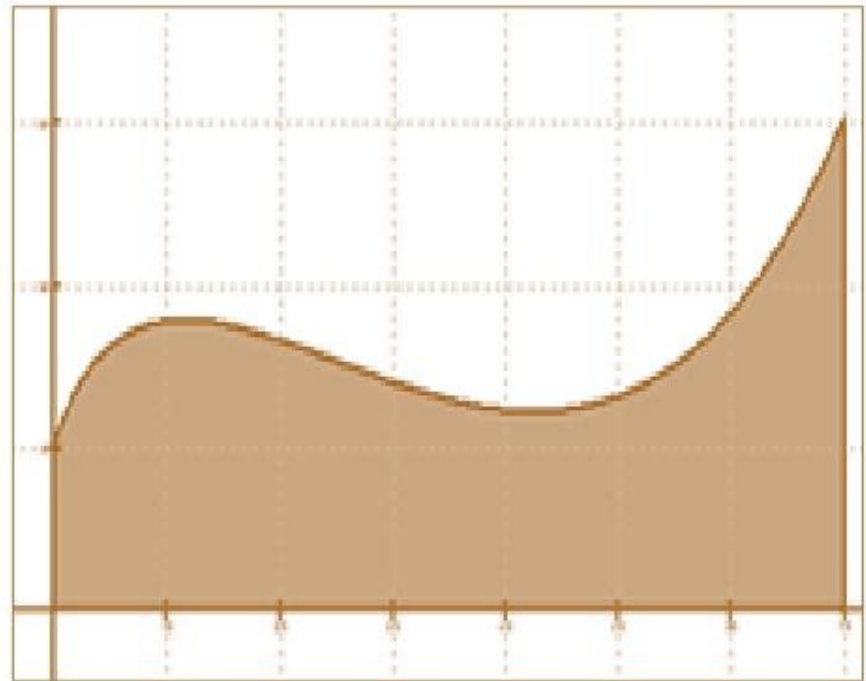


Integral Calculus

- This branch of calculus focuses on integration/accumulation.



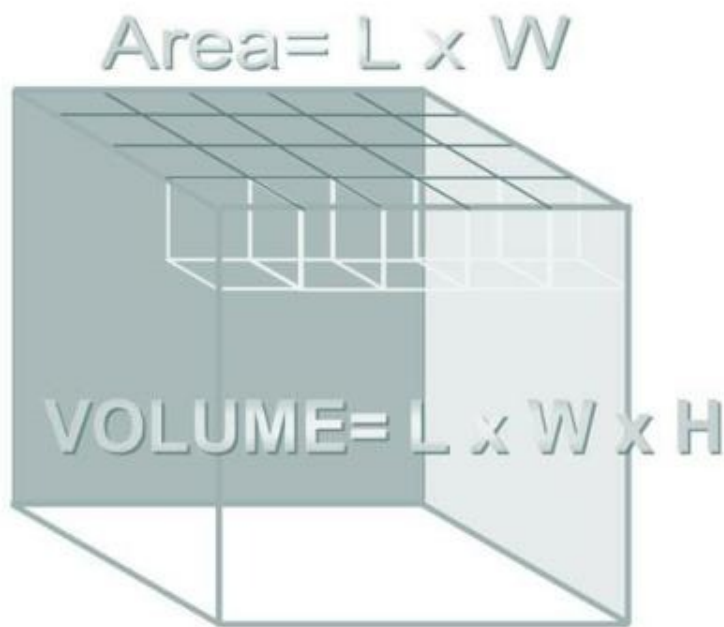
You could easily find the area of this rectangle using simple algebra.



To find the area of this curved region, you need to use Calculus.

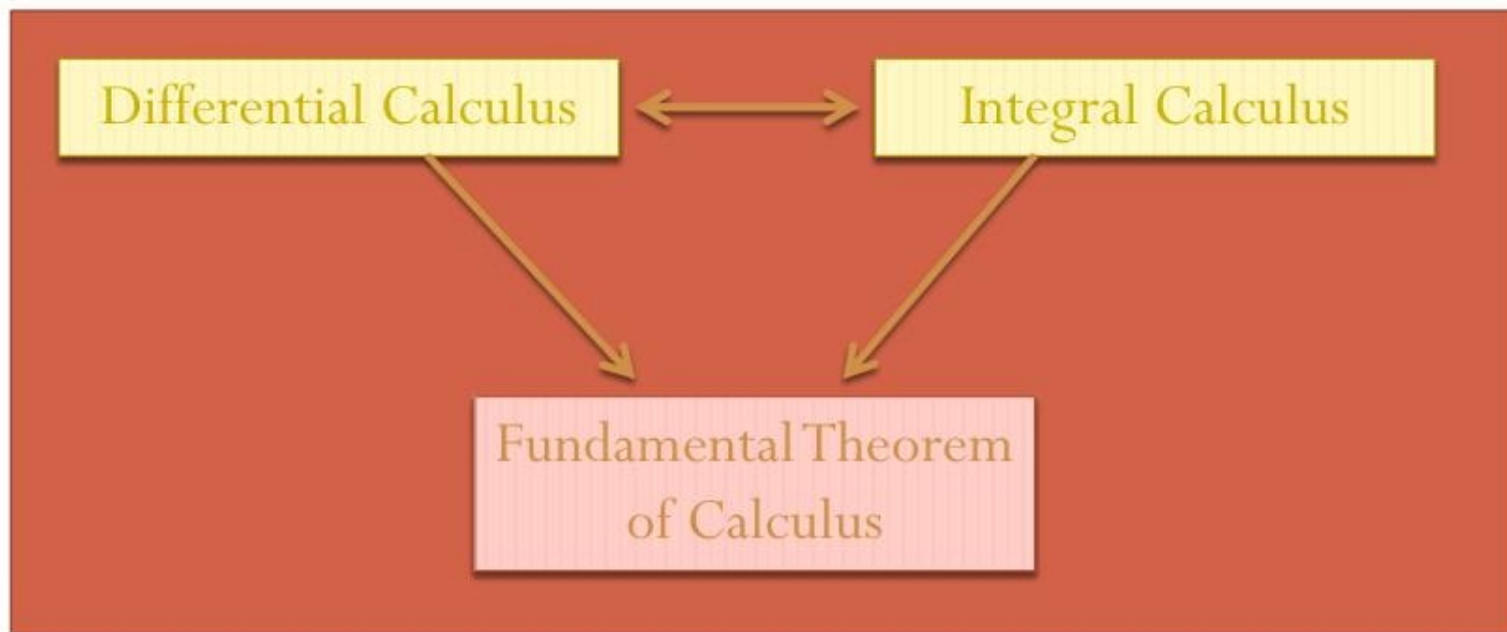
Integral Calculus: Real World Applications

- Integral Calculus is also used in fields such as science, economics, and engineering.
- Integral Calculus is useful in solving problems involving:
 - Area
 - Volume



Connecting Differential and Integral Calculus

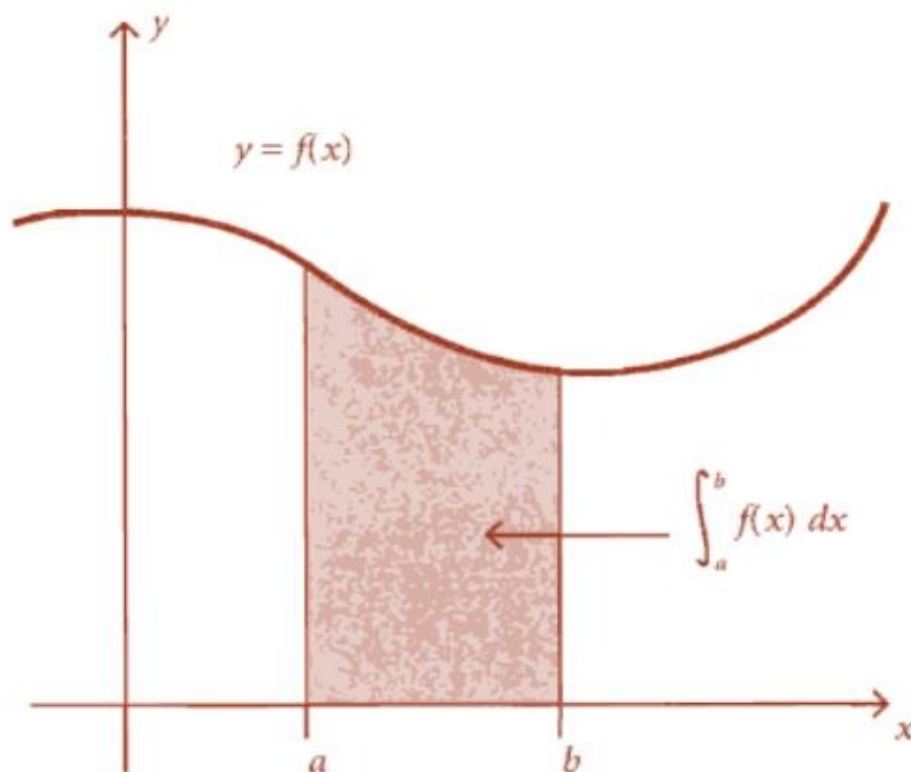
- Differential Calculus is deals with limits and derivatives.
- Integral Calculus deals with integrals.
- The two branches are connected by the Fundamental Theorem of Calculus.



Fundamental Theorem of Calculus

- The Fundamental Theorem of Calculus proves the following:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x) \quad \text{so} \quad f'(x) = f'(x)$$



Who Discovered Calculus?

- The answer isn't one person but two: Sir Isaac Newton and Gottfried Wilhelm Leibniz. Each from two different places, they both discovered calculus at different times.

Sir Isaac Newton

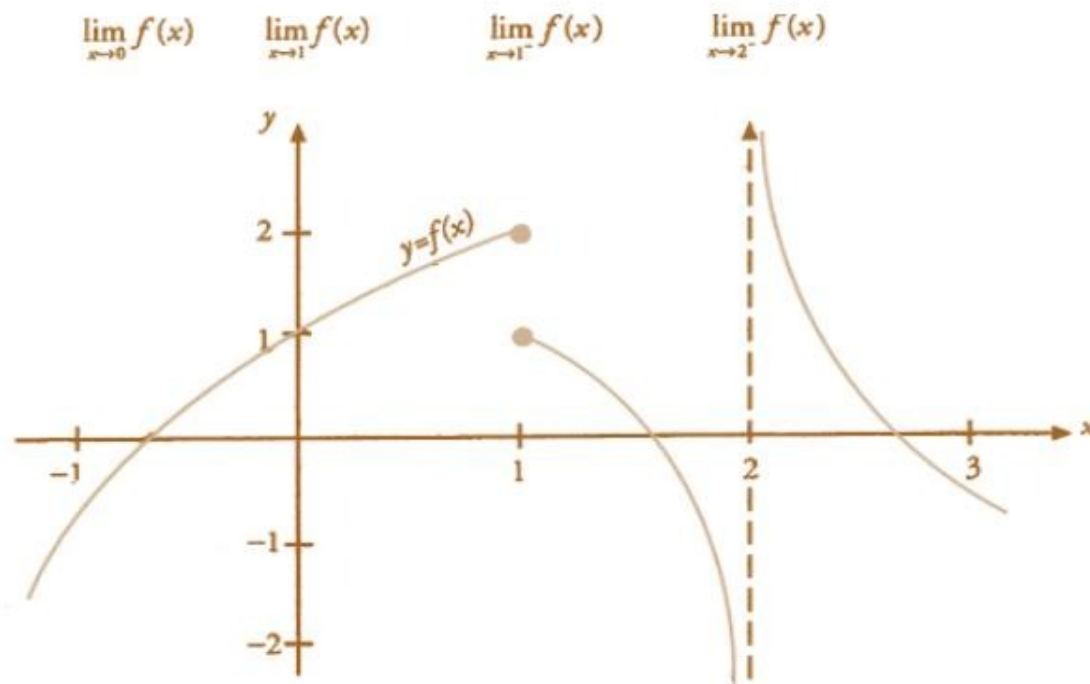
- Discovered Calculus between 1665 and 1667.
- Did not immediately publish his findings- this led to a controversy over who discovered Calculus first.

Gottfried Wilhelm Leibniz

- Discovered Calculus 8 years after Newton.
- Was very open about his findings.
- Credited with discovering the dy/dx notation, the integral symbol, and the equal sign.

Section 1.1: Limits and Continuity

- Limits are used to describe functions with specific properties.
- The limit is the fundamental notion of calculus.
- The limit is the value that x approaches.



Solving Basic Problems with Limits

1. Evaluate $\lim_{x \rightarrow 3} \frac{x-1}{x^2+2x-3}$

Step 1: Substitute 3 for x.

$$\lim_{x \rightarrow 3} \frac{(3)-1}{(3)^2+2(3)-3} = \frac{2}{12} = \frac{1}{6}$$

A Limit Where Two Factors Cancel

1. Evaluate $\lim_{x \rightarrow -3} \frac{3x+9}{x^2-9}$

Step 1: Substitute -3 for x.

$$\lim_{x \rightarrow -3} \frac{3(-3)+9}{(3)^2-9} = \frac{0}{0}$$

You can not have 0/0, so you must do something extra.

Step 2: Factor out the top and bottom and simplify.

$$\lim_{x \rightarrow -3} \frac{3(x+3)}{(x+3)(x-3)} = \frac{3}{(x-3)}$$

Step 3: Again, substitute -3 for x, and evaluate.

$$\lim_{x \rightarrow -3} \frac{3}{[(-3)-3]} = \frac{3}{-9} = \frac{-1}{2}$$

A Limit That Does Not Exist

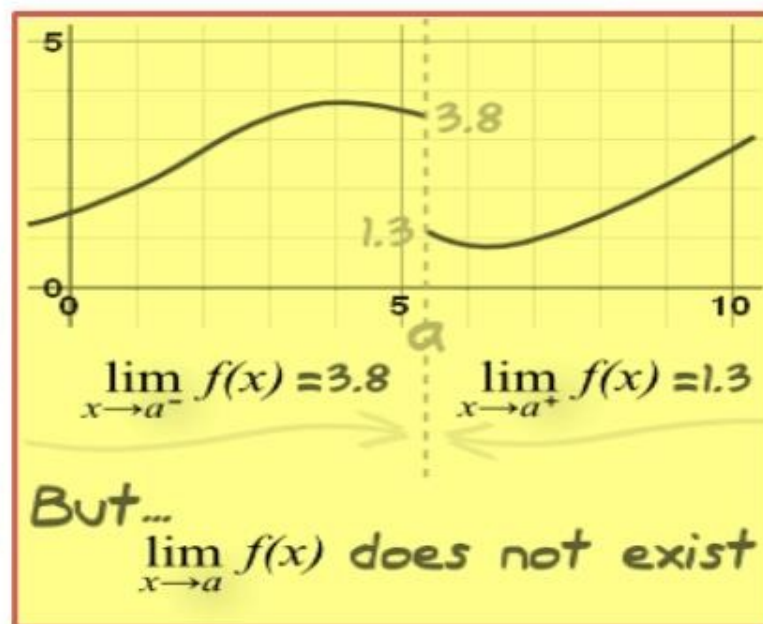
1. Evaluate $\lim_{x \rightarrow 0} \frac{1}{x}$

Step 1: Substitute 0 for x.

$$\lim_{x \rightarrow 0} \frac{1}{(0)} = \frac{1}{0} = \text{Limit does not exist.}$$

When you have a number over zero, it means the limit does not exist.

Graph of a function where the limit does not exist.

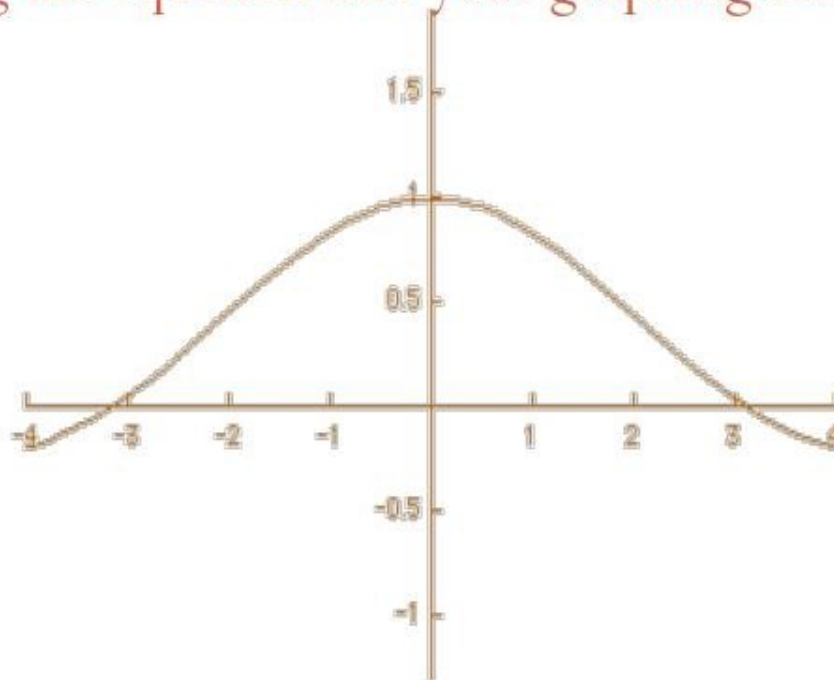


Approximating the Value of a Limit

1. Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

- This equation can not be simplified with algebra.
- However, you can still draw a graph and look at the table of values.

Step 1: Plug the equation into your graphing calculator.



Approximating the Value of a Limit

Step 2: Use your graphing calculator to view the table of values for the function.

$$\lim_{x \rightarrow 0^+}$$

X	$\frac{\sin x}{x}$
0.1	0.998334
0.01	0.999983
0.001	0.99999983
0.0001	0.999999983
0.00001	0.9999999983

$$\lim_{x \rightarrow 0^-}$$

x	$\frac{\sin x}{x}$
-0.1	0.998334
-0.01	0.999983
-0.001	0.99999983
-0.0001	0.999999983
-0.00001	0.9999999983

Step 3: Look at the table and approximate the value for the limit.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

A Case Where One-Sided Limits are Needed

1. Evaluate $\lim_{x \rightarrow 0} \frac{x}{|x|} \quad \left\{ \begin{array}{l} x > 0 = x \\ x < 0 = -x \end{array} \right\}$

Step 1: Since $|x| = x, x > 0$ Step 3: Since $x = -x, x < 0$

$$\lim_{x \rightarrow 0^+} \frac{x}{x}$$

$$\lim_{x \rightarrow 0^+} \frac{x}{-x}$$

Step 2: Simplify

$$\lim_{x \rightarrow 0^+} \frac{\cancel{x}}{\cancel{x}} = 1$$

Step 4: Simplify

$$\lim_{x \rightarrow 0^+} \frac{x}{-x}$$

Therefore, $\frac{x}{|x|} \quad \left\{ \begin{array}{l} 1, x > 0 \\ -1, x < 0 \end{array} \right\}$

A Limit Where Two Factors Cancel

1. Evaluate $\lim_{x \rightarrow -3} \frac{3x+9}{x^2-9}$

Step 1: Substitute -3 for x.

$$\lim_{x \rightarrow -3} \frac{3(-3)+9}{(3)^2-9} = \frac{0}{0}$$

You can not have 0/0, so you must do something extra.

Step 2: Factor out the top and bottom and simplify.

$$\lim_{x \rightarrow -3} \frac{3(x+3)}{(x+3)(x-3)} = \frac{3}{(x-3)}$$

Step 3: Again, substitute -3 for x, and evaluate.

$$\lim_{x \rightarrow -3} \frac{3}{[(-3)-3]} = \frac{3}{-9} = \frac{-1}{2}$$

A Limit Involving More Than One Variable

1. Evaluate $\lim_{b \rightarrow -x} \frac{b^2 - x^2}{b + x}$

Step 1: Substitute $-x$ for b .

$$\frac{(-x)^2 - x^2}{(-x) + x} = \frac{0}{0}$$

Step 2: Factor out the numerator.

$$\frac{(b-x)(\cancel{b+x})}{\cancel{-b+x}} = b-x$$

Step 3: Substitute $-x$ for b .

$$(-x)-x = -2x$$

Section 1.2: Computation of Limits

- The computation of limits can be done by applying knowledge of basic theorems.
- According to Webster's dictionary, theorems are “a formula, proposition, or statement in mathematics or logic deduced or to be deduced from other formulas or propositions”.

Theorem Description	Formula
Theorem 2.1: The constant c stays the same.	For any c and real number a , $\lim_{x \rightarrow a} c = c$
Theorem 2.2: The value to which x approaches, a , can be substituted for x .	For any real number a , $\lim_{x \rightarrow a} x = a$

Theorems

Theorem Description	Formula
Theorem 2.3: i) When multiplying a constant with the function, the constant remains the same and the value for x is applied to the function. ii) When two functions are being added, the limit applies to both. iii) When two functions are being multiplied, the limit applies to both. iv) When two functions are being divided, the limit applies to both, if the limit for the denominator does not equal zero. In that case, the limit does not exist.	i) $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$ ii) $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$ iii) $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = [\lim_{x \rightarrow a} f(x)] \cdot [\lim_{x \rightarrow a} g(x)]$ iv) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$ (if $\lim_{x \rightarrow a} g(x)$ is not 0)

Corollaries

- Applying part (iii) of Theorem 2.3 with $g(x)=f(x)$ will result in a corollary.
- In mathematics, a corollary is the result following immediately from another result. A corollary is derived from a theorem.

Corollary Description	Formula
Corollary 2.1 When a function is being squared, it can be written as that function multiplied by the same function, and the limit applies to both.	Suppose that $\lim_{x \rightarrow a} f(x)$ exists. Then, $\lim_{x \rightarrow a} [f(x)^2] = \lim_{x \rightarrow a} [f(x) \cdot f(x)] =$ $\left[\lim_{x \rightarrow a} f(x) \right] \left[\lim_{x \rightarrow a} f(x) \right]$
Corollary 2.2 For a function in which x is brought to a power, the value for which the limit approaches can be substituted for x and brought to the same power.	For any integer $n > 0$ and any real number a, $\lim_{x \rightarrow a} x^n = a^n$

Find the Limit of a Power Function

1. Evaluate $\lim_{x \rightarrow -2} x^5$

Step 1: Apply Corollary 2.2 to evaluate.

$$\lim_{x \rightarrow -2} x^5 = (-2)^5 = -32$$

Finding the Limit of a Polynomial

1. Apply the rules of limits to evaluate $\lim_{x \rightarrow 2} (3x^2 - 5x + 4)$

Step 1: Use Theorem 2.3 (ii).

$$\lim_{x \rightarrow 2} (3x^2) - \lim_{x \rightarrow 2} (5x) + \lim_{x \rightarrow 2} (4)$$

Step 2: Use Theorem 2.3 (i)

$$3 \lim_{x \rightarrow 2} x^2 - 5 \lim_{x \rightarrow 2} x + 4$$

Step 3: Use Corollary 2.2

$$3(2)^2 - 5(2) + 4 = 6$$

Basically, use the knowledge we know for calculus to solve. The constant stays the same, and then you can substitute the value for x .

Finding the Limit of a Rational Function

1. Apply the rules of limits to evaluate $\lim_{x \rightarrow 3} \frac{x^3 - 5x + 4}{x^2 - 2}$

Step 1: Use Theorem 2.3 (iv).

$$\frac{\lim_{x \rightarrow 3} \frac{x^3 - 5x + 4}{\lim_{x \rightarrow 3} x^2 - 2}}$$

Step 3: Use Corollary 2.2.

$$\frac{(3)^3 - 5(3) + 4}{(3)^2 - 2} = \frac{16}{7}$$

Step 2: Use Theorem 2.3 (i)
and Theorem 2.3(ii).

$$\frac{\lim_{x \rightarrow 3} x^3 - \lim_{x \rightarrow 3} 5x + \lim_{x \rightarrow 3} 4}{\lim_{x \rightarrow 3} x^2 - \lim_{x \rightarrow 3} 2}$$

A Limit Where Two Factors Cancel

1. Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{1 - x}$

Step 1: Substitute 1 for x.

$$\lim_{x \rightarrow 1} \frac{(1)^2 - 1}{(1) - 0} = \frac{0}{0}$$

Remember, one of our theorems stated that the denominator can not equal zero. To solve this problem, you will need to factor.

Step 2: Factor out the numerator and factor a -1 from the denominator in order to cancel.

$$\lim_{x \rightarrow -3} \frac{\cancel{(x-1)}(x+1)}{\cancel{-(x-1)}-1} = \frac{x+1}{-1}$$

Step 3: Substitute 1 for x.

$$\frac{1+1}{-1} = -2$$

Finding Limits by Applying the Sum or Difference of Cubes

1. Evaluate $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 + 2x - 3}$

Step 1: Substitute 1 for x.

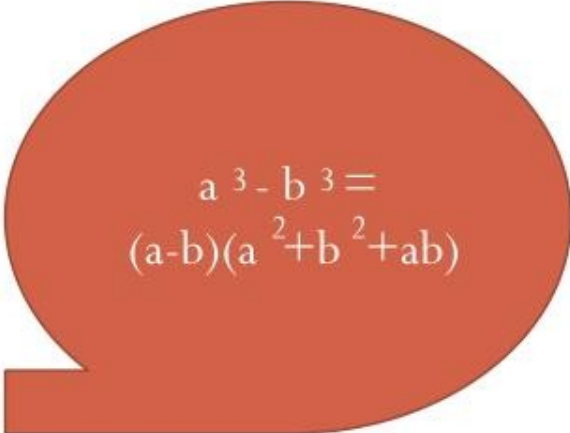
$$\frac{(1)^3 - 1}{(1)^2 + 2(1) - 3} = \frac{0}{0}$$

Step 2: Since this is in 0/0 form, we will need to do something extra. We can apply the difference of cubes formula for the numerator and factor out the denominator.

$$\frac{(x-1)(x^2 + x + 1)}{(x+3)(x-1)}$$

Step 3: Substitute 1 for x to get an answer this time.

$$\frac{1+1+1}{1+3} = \frac{3}{4}$$


$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Theorems

Theorem Description	Formula
Theorem 2.4: You can simply substitute a value a into x for the function.	For any polynomial $p(x)$ and any real number a, $\lim_{x \rightarrow a} p(x) = p(a)$
Theorem 2.5: You may bring limits “inside” the nth roots, and then use existing rules for computing the limit inside.	Suppose that $\lim f(x) = L$ and n is any positive integer. Then, $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim f(x)} = \sqrt[n]{L}$

Evaluating the Limit of an nth Root of a Polynomial

1. Evaluate $\lim_{x \rightarrow 2} \sqrt[5]{3x^2 - 2x}$

Step 1: Apply Theorems 2.4 and 2.5 to solve.

$$\sqrt[5]{\lim_{x \rightarrow 2} (3x^2 - 2x)} = \sqrt[5]{8}$$

Theorem 2.5 tells us we can apply the limits inside the nth root. Theorem 2.4 tells us we can substitute the value and evaluate.

Theorems

Theorem Description	Formula
Theorem 2.6 Says that limits of the sine, cosine, exponential and natural logarithm functions are found simply by substitution.	i) $\lim_{x \rightarrow a} \sin x = \sin a$ ii) $\lim_{x \rightarrow a} \cos x = \cos a$ iii) $\lim_{x \rightarrow a} e^x = e^a$ iv) $\lim_{x \rightarrow a} \ln x = \ln a$

Finding the Limit of a Quotient of Two Trigonometric Functions

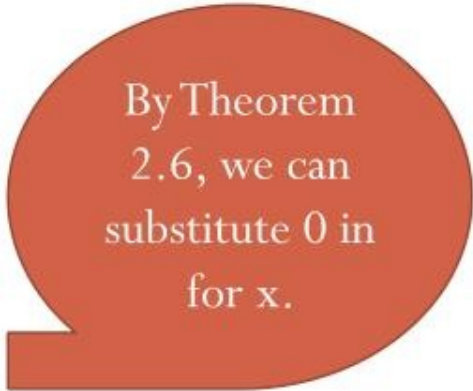
1. Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{\cos x}$

Step 1: Substitute 0 for x.

$$\lim_{x \rightarrow 0} \frac{\sin(0)}{\cos(0)}$$

Step 2: Evaluate

$$\frac{0}{1} = 0$$



By Theorem 2.6, we can substitute 0 in for x.

More Functions with Trigonometric Limits

Ex. Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\tan x}$

Step 1: Substitute 0 in for x.

$$\frac{1 - \cos(0)}{\tan(0)} = \frac{0}{0}$$

Step 2: Since it is in this form, we must rewrite using different formulas and simplify.

$$\frac{2 \sin^2 x/2}{\tan x} = \frac{2 \sin x/2 \sin x/2}{\tan x} = \frac{2x^2}{x} = 2x$$

Step 3: Substitute 0 for x.

$$2(0) = 0$$

Formulas:

$$1 - \cos x = 2 \sin^2 x/2$$

$$\frac{\sin x}{x} = x$$

Finding One-Sided Limits of a Function

Ex. $\lim_{x \rightarrow 2} f(x)$ where $f(x) = \begin{cases} 3x^2 - 2x + 1 & x < 2 \\ x^3 + 1 & x \geq 2 \end{cases}$

Step 1: Solve for the left hand limit. In this case, the LHL of $f(x)$ would be the function in which x is less than 2, so x approaches 2-.

$$\text{LHL } \lim_{x \rightarrow 2^-} 3x^2 - 2x + 1$$

Step 2: Change the limit to h approaches zero, and plug in $(2-h)$ for x .

$$\lim_{h \rightarrow 0} 3(2-h)^2 - (2-h) + 1$$

Step 3: Substitute 0 for h and simplify.

$$3(2-0)^2 - 2(2-0) + 1 = 9$$

Finding One-Sided Limits of a Function

Cont.

Step 4: Solve for the right hand limit. In this case, the RHL of $f(x)$ would be the function in which x is greater than or equal to 2, so x approaches $2+$.

$$\text{RHL } \lim_{x \rightarrow 2+} x^3 + 1$$

Step 2: Change the limit to h approaches zero, and plug in $(2+h)$ for x .

$$\lim_{h \rightarrow 0} (2+h)^3 + 1$$

Step 3: Substitute 0 for h and simplify.

$$(2+0)^3 + 1 = 9$$

LHL=RHL, so limit $x \rightarrow 2$ $f(x)$ exists.

Calculus in Real Life: A Physics-Related Problem

Suppose that the position function for an object is given by $f(t) = t^2 + 2$ where t is measured in seconds. Find the instantaneous velocity of the object at time $t=1$.

Step 1: Apply the formula, and foil.

$$\lim_{h \rightarrow 0} \frac{f(1+h^2+2) - 3}{h} = \frac{(1+2h+h^2) - 1}{h}$$

Step 2: Simplify.

$$\frac{(2h+h^2)}{h} = \frac{h(2+h)}{h} = 2+h$$

Step 3: Now substitute the limit 0 in for x .

$$2+0=2$$

So, the instantaneous velocity of this object at time $t=1$ is 2 feet per second.

Instantaneous velocity of an object at time $t=1$ is given by the limit:

$$\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

Section 1.3: Continuity and Its Consequences

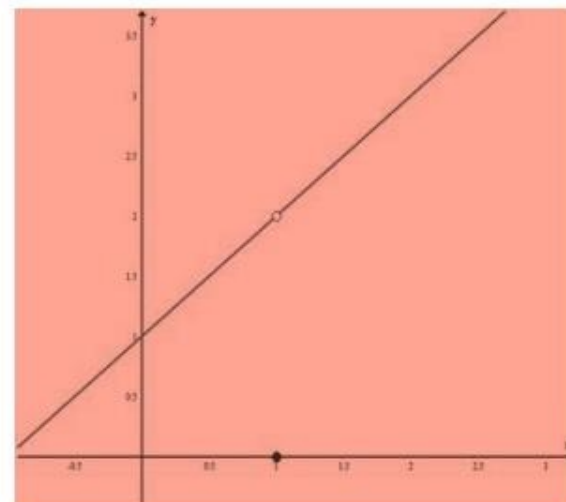
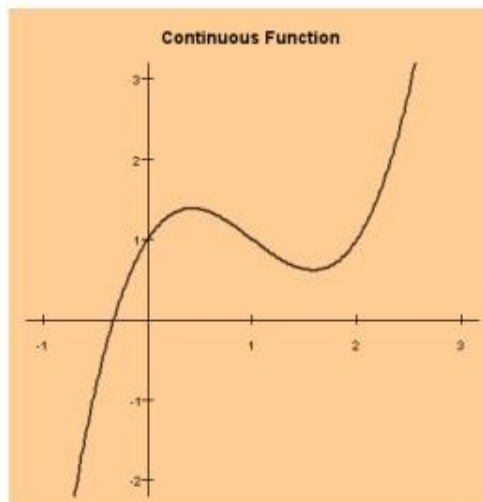
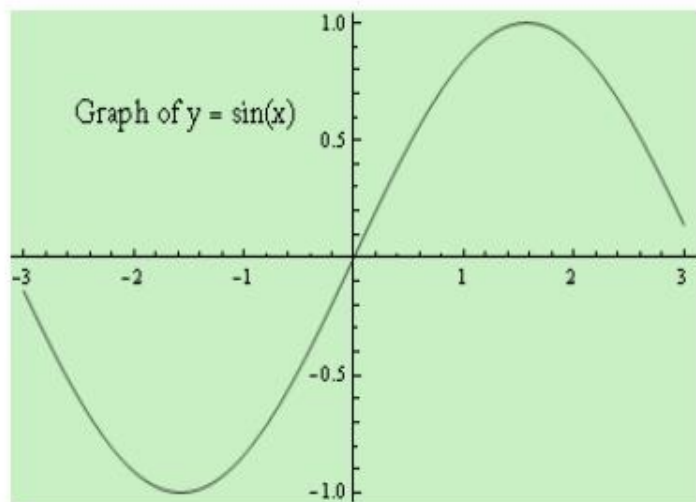
- A function is said to be continuous on an interval if its graph on that interval can be drawn without interruption.

Definition:

A function is continuous at a point exactly when you can compute its limit at that point by simply substituting.

A function f is said to be continuous at $x=a$ when

- i) $f(a)$ is defined
- ii) $\lim_{x \rightarrow a} f(x)$ exists
- iii) $\lim_{x \rightarrow a} f(x) = f(a)$



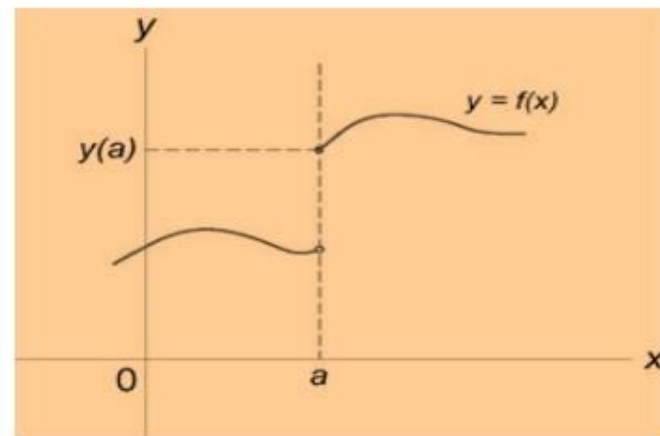
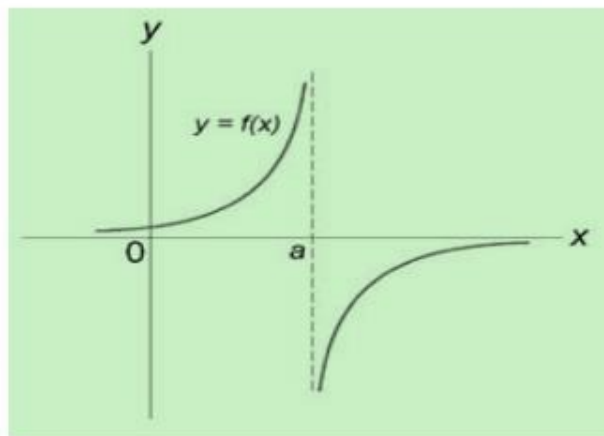
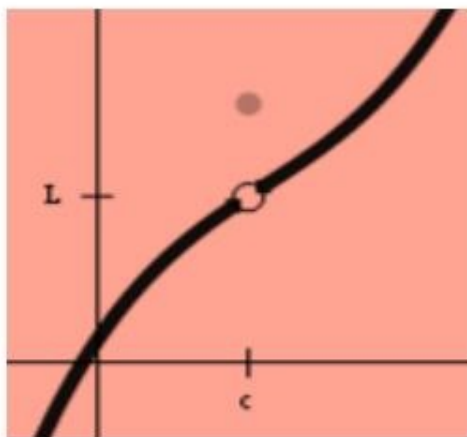
Discontinuous Functions

Definition:

- i) The graph has a whole at $x=a$.
- ii) The graph has a jump at $x=a$.
- iii) The graph has a jump at $x=a$.
- iv) The graph “blows up” at $x=a$.

A function f is said to be discontinuous when:

- i) $f(a)$ is not defined
- ii) $f(a)$ is defined but $\lim_{x \rightarrow a} f(x)$ does not exist.
- iii) $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} f(a)$ is defined, but $\lim_{x \rightarrow a} f(x)$ does not equal $f(a)$.
- iv) $\lim_{x \rightarrow a} f(x)$ does not exist.



Finding Where a Rational Function is Continuous

1. Determine where $f(x) = \frac{x^2 + 2x - 3}{x - 1}$

Step 1: Factor the numerator and cancel where possible.

$$\frac{\cancel{(x-1)}(x+3)}{\cancel{(x-1)}} = (x+3)$$

This answer says that graph of f is a straight line, but with a hole in it at $x=1$. So, f is discontinuous at $x=1$, but continuous elsewhere.

Discussing the Continuity of a Function by Finding the Left Hand and Right Hand Limit

- A function is said to be continuous if the left hand limit is equal to the right limit is equal to the value of the function.

Ex. Discuss the continuity of a function.

$$f(x) = x^2 \quad x < 2$$

$$x \rightarrow 2 \quad 3x - 1 \quad x \geq 2$$

$$\text{LHL } f(x) = (2-0)^2 = 4$$

$$\text{RHL } f(x) = 3(2+0) - 1 = 5$$

LHL does not equal RHL, so $f(x)$ at $x \rightarrow 2$ is not continuous.

Discussing the Continuity of a Function by Finding the Left Hand and Right Hand Limit

$$\begin{array}{lll} \text{Ex. } f(x) = & x^2 & x < 2 \\ & 3 & x = 2 \\ & 3x - 3 & x > 2 \end{array}$$

$$\text{LHL } f(x) = (2-0)^2 = 4$$

$$\text{RHL } f(x) = 3(2+0) - 3 = 3$$

LHL does not equal RHL, so the function is not continuous.

Finding Continuity Involving Absolute Value

Ex. State whether the function is continuous or not.

$$f(x) = \frac{|x-9|}{x-9} \quad x=9 \quad \begin{array}{l} \text{LHL: } -(x-9) \\ \text{RHL: } x-9 \end{array}$$

Step 1: Solve for both the left and right hand limits and simplify.

LHL

$$\lim_{x \rightarrow 9^-} \frac{-(x-9)}{x-9} = -1$$

RHL

$$\lim_{x \rightarrow 9^+} \frac{x-9}{x-9} = 1$$

LHL does not equal RHL, so the function is not continuous.

Finding a and b so the Function is Continuous at Every Point

Ex. Find a and b so that $f(x)$ is continuous at every point.

$$f(x) = \begin{cases} -12, & x < -1 \\ ax+b, & -1 < x < 2 \\ 6, & x > 2 \end{cases}$$

Step 1: Start out by finding the LHL for x approaches -1 . In order to do this, you must rewrite it so that the limit is changed to h approaches 0 and substitute 0 for h .

LHL

$$\lim_{h \rightarrow 0} a(-1+h)+b = -a+b$$

Step 2: Find the RHL the same way.

RHL

$$\lim_{h \rightarrow 0} -12$$

Step 3: Set the LHL and RHL equal to each other.

$$-a+b = -12$$

Finding a and b so the Function is Continuous at Every Point Cont.

Step 4: Find the LHL for x approaches 2.

LHL

$$\lim_{h \rightarrow 0} a(2+h)+b = 2a+b$$

$$h \rightarrow 0$$

Step 4: Find the RHL for x approaches 2.

RHL

$$\lim_{h \rightarrow 0} 6$$

$$h \rightarrow 0$$

Step 5: Set the LHL and RHL equal to each other.

$$2a+b=6$$

Finding a and b so the Function is Continuous at Every Point Cont.

$$-a+b=-12$$

$$2a+b=6$$

Step 6: Now use substitution to solve for a and b.

$$2a+(-12+a)=6$$

$$3a-12=6$$

$$3a=18$$

$$a=6$$

$$-a+b=-12$$

$$b=-12+a$$

$$-(6)+b=-12$$

$$b=-6$$

Section 1.4: Limits Involving Infinity

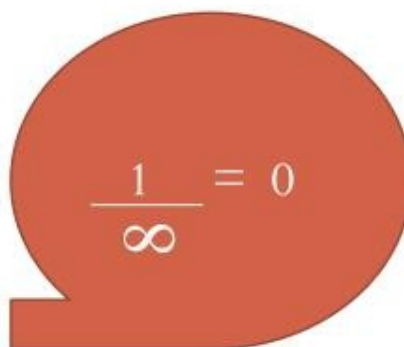
$$\text{Ex. } \lim_{x \rightarrow \infty} \frac{4x}{x-14}$$

Step 1: Divide the numerator and denominator by the highest power of x . In this problem, the highest power is simply 1, so divide each part by x and simplify.

$$\lim_{x \rightarrow \infty} \frac{\frac{4\cancel{x}}{\cancel{x}}}{\frac{\cancel{x}}{\cancel{x}} - \frac{14}{x}} = \frac{4}{1 - \frac{14}{x}}$$

Step 2: Substitute infinity in for x .

$$\frac{4}{1 - \frac{14}{\infty}} = \frac{4}{1 - 0} = 4$$


$$\frac{1}{\infty} = 0$$

More Problems Involving Infinity

$$\text{Ex. } \lim_{x \rightarrow \infty} \frac{x^2 - 3x + 13}{x^3 + 5x^2 + 8}$$

Step 1: Divide the numerator and denominator by the highest power of x . The highest value is x^3 and replace each x with infinity and simplify.

$$\frac{\frac{x^2}{x^3} - \frac{3x}{x^3} + \frac{13}{x^3}}{\frac{\cancel{x^3}}{x^3} + \frac{5x^2}{x^3} + \frac{8}{x^3}}$$

More Problems Involving Infinity

Step 2: Since the limit approaches infinity, you can also substitute infinity in for x.

$$\frac{\frac{x^2}{\infty} + \frac{3x}{\infty} + \frac{13}{\infty}}{0 + \frac{5x}{\infty} + \frac{8}{\infty}} = \frac{0}{1} = 0$$



Pop Quiz Question: Section 1

What is the first step in solving questions with limits?

- a) Factoring out the numerator or denominator of the function
- b) Graphing the function
- c) Substituting the value for x into the function
- d) Changing the $\frac{0}{0}$ form

The correct answer is b. You must first substitute the value for x . From there, you can tell if a function is in $\frac{0}{0}$ form and requires more steps for solving.



For example...

In the problem $\lim_{x \rightarrow 2} \frac{x-1}{x}$

the first thing we would do is plug 2 in for x. Then, we would get

$$\frac{2-1}{2} = \frac{1}{2}$$



Pop Quiz Question: Section 2

How are theorems used in the computation of limits?

- a) Theorems are the organized proofs for limits
- b) Theorems provide us with formulas for solving the problem
- c) Theorems are the scientists who have designed methods for computing limits
- d) Theorems tells us when a function is in $\frac{0}{0}$ form

The correct answer is b. Theorems are formulas derived from other formulas. We can use them to make computing limits a possible and easy task.



For example...

Theorem 2.4 tells us that for any polynomial $p(x)$ and any real number a ,

$$\lim_{x \rightarrow a} p(x) = p(a)$$

$x \rightarrow a$

This tells us that we can simply substitute the value of a into x for the function, making it an easy problem to solve.

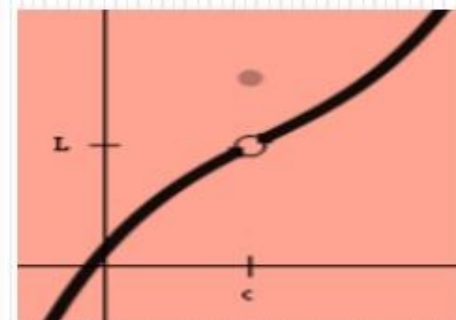


Pop Quiz Question: Section 3

True or False:

This graph is a continuous function.

The correct answer is False.



Bonus: What type of discontinuous function is this? How is it discontinuous?

It has a hole, or in other words, $f(x)$ is not defined.



Pop Quiz Question: Section 4

True or False:

The first step in solving problems with limits of infinity is to divide by the lowest power of x .

The correct answer is False. You should divide by the highest power of x .



For example...

In the problem $\lim_{x \rightarrow \infty} \frac{5x - 3}{x^2 - 3}$

the highest power is 2, so we must divide each part by x^2

Summary of Objectives

- Calculus is a branch of mathematics divided into differential calculus and integral calculus.
- The basic concept that must be understood in calculus is the limit.
- Limits are values that x approaches.
- When solving problems with limits, it is necessary to first substitute the value of x into the problem. From there, you can apply your knowledge of theorems and trigonometric formulas to evaluate and simplify.
- Some functions are continuous, while others are discontinuous. A continuous function is a steady, smooth line. Discontinuous functions can have holes, breaks, or jumps- they are not smooth.
- There are also left hand and right hand limits. These require you to take the value of x to the $-$ and $+$ signs: basically solve to the left and right of each value.
- Concluding the chapter is limits with infinity. They require you to divide each part by the highest power of x .